

USE-CASE FLOW SPECIFICATION

System: Warehouse Inventory & Pallet Location Tracker

Use Case: UC-02 – Assign Pallet Location

1. Preconditions

- Warehouse Operator is authorized to use the system.
- The pallet has a valid pallet ID/SKU.
- The target 3D shelf position is available in the system.
- Rack capacity information is available.

2. Postconditions

- The pallet is assigned to the selected 3D shelf position.
- The pallet location is updated in the system.
- The stock movement is recorded.

3. Main Success Scenario

1. Warehouse Operator scans or enters the pallet ID.
2. System identifies the pallet.
3. Warehouse Operator selects a 3D shelf position.
4. System validates the selected shelf coordinates.
5. System checks the rack's current load capacity.
6. System confirms that the rack can accommodate the pallet.
7. System assigns the pallet to the selected shelf position.
8. System updates the pallet's location.
9. System records the stock movement.
10. System displays a successful storage confirmation.

4. Alternate Flow – Rack Capacity Exceeded

1. At Step 6, the system determines that adding the pallet would exceed the rack's maximum weight limit.
2. System rejects the proposed pallet placement.
3. System displays a message indicating that the rack capacity has been exceeded.
4. Warehouse Operator selects another valid shelf position.
5. The flow resumes from Step 4.

